ASSIGNMENT 2

Investment memo

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FlowAssist — Agent-assist copilot for B2B payments

Data & AI Product: From Diagnosis to Delivery · Albert School / Mines Paris-PSL
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PDF edition · 6 November 2026 · Coursework in English, following the supplied briefs.

CONDITIONAL GO if paired active handling-time saving is **≥41.2%** on pre-labelled eligible tracing cases, grounding-check precision **≥98%**, QA **≥92%** and critical exposures **zero**; independently validate before any production investment.

How much?

Year-1 base net value is **€141,340**, on 144,000 annual tickets at €10/ticket. The three-category S2 model yields €174,892 gross time-equivalent value, less €19,848 expected errors, €704 inference/retries and €78,000 fixed/semi-fixed costs, plus €65,000 SLA/retention value. Base adoption is **15% / 35% / 60% / 75% by quarter**, averaging 46.25%. Integration is €60,000 upfront, amortised at €20,000/year; reported net value is an annualised model, not year-1 cash profit.

€/year	Pessimistic	Base	Optimistic
Net annual value	€-59,828	€141,340	€434,862
Probability	20%	60%	20%
Do nothing	€32,130	€57,600	€96,600
Next-best project	€100,000	€125,000	€145,000

How sure?

Expected value is **€159,810**, +€34,810 versus the base hurdle and +€102,210 versus base do-nothing. The dominant guess is 48% time saving on eligible tracing cases. Workbook sensitivity gives €98,251 at 30% and €134,158 at 45%; the interpolated break-even is **41.174%**, rounded up to **41.2%**. Test using paired manual/assisted timing on a frozen 200-case benchmark, including review and fallback time. These are targets, not measured results.

What if we are wrong?

The pessimistic case loses **€59,828/year**. All seven cost lines are included; financial/data disclosure tail risks remain unpriced beyond operational expected losses. CCO/DPO own the regulatory tail, Ops owns answer quality, Engineering owns deterministic isolation and blocking. A second model cannot guarantee a safe transaction match. Stop drafting on a critical exposure. No real company data or human-validated performance has been supplied.

Compared to what?

Do nothing assumes 4% natural efficiency gain: **€57,600/year**. The named alternative is the **Merchant Self-Serve Dispute & Reconciliation Portal**, worth **€125,000/year** in Base. The S3 tracing-only MVP does not establish dispute/compliance savings or all €65,000 strategic benefits; a timing pass permits further validation, not the whole business case.

Build, buy or orchestrate?

Orchestrate: weighted scores Build 3.15 / Buy 2.15 / Orchestrate 4.35. Retain permission checks and private read-only retrieval; evaluate hosted model components. At the workbook's €0.92/outcome assumption, vendor usage is €16,090 versus €50,204 comparable internal cost; integration, liability and fit decide. Switch to Buy if an audited suitable connector, approved data terms and liability terms cost <€0.40/outcome. Vendor/API prices rechecked 6 November; USD list prices and workbook EUR assumptions are distinguished in the annex.

Source: 02_sizing.xlsx, Inputs / Upside / Decision / Build or buy. Formula outputs recomputed from inputs. Original Excel and full assumption register accompany this PDF.

Model and assumption register

Formula outputs below are recomputed from the original workbook; the editable Excel is retained. All internal-data “measured” claims are educational assumptions, not supplied observations. Base category shares total 100%; Low totals 94% and High 106%, so those scenarios need reconciliation before investment. Retention is €30k judgement; the note's multiplication gives €28,800. Inference/errors are charged to time-equivalents in the original template, not all assisted calls; at 54,654 eligible assisted Base conversations, inference would be about €2,200 rather than €704. Oversight is additional to review time. Keep these historical formulas for transparency, not as an audited forecast.

Rollout remains conditional: Q1 identity/no-send gates; Q2 QA ≥92%, grounding precision ≥98%, zero critical exposures; Q3 ≥41.2% paired saving plus independent review; Q4 continued guardrails. Integration is amortised, so annual net value is not year-1 cash flow. Do not double-count staffing avoidance.

02 · Scenario economics

Year 1	Pessimistic	Base	Optimistic
Adoption	35.00%	46.25%	55.00%
Gross time-equivalent value	€49,630	€174,892	€411,855
Expected operational errors	€19,594	€19,848	€17,427
Inference + retries	€365	€704	€967
Other value	€27,000	€65,000	€95,000
Fixed/semi-fixed	€116,500	€78,000	€53,600
Net annual value	€-59,828	€141,340	€434,862
Do nothing	€32,130	€57,600	€96,600
Named hurdle	€100,000	€125,000	€145,000

Expected value: 0.2 × (€-59,828) + 0.6 × €141,340 + 0.2 × €434,862 = **€159,810**. Probabilities sum to 1.00. The Decision tab compares EV to the Base do-nothing and Base hurdle. Probability-weighted comparisons are distinct: do-nothing €60,306; hurdle €124,000.

Base category	Equivalent tickets	Gross value	Errors	AI + retry	Net variable
Tracing	11,988.00	€119,880	€4,496	€483	€114,902
Disputes	3,836.16	€38,362	€10,358	€154	€27,850
KYC / holds	1,665.00	€16,650	€4,995	€67	€11,588

03 · Seven cost lines and sensitivity

Cost line	Base €/year	Basis / owner
Tokens and inference	€704	€0.035 × 1.15 × 17,489.16 equivalents; Engineering
Integration	€20,000	€60,000 / 3 years; Engineering
Evaluation	€22,000	Dataset + graders; Ops QA
Supervision	€16,500	0.3 FTE × €55,000; Ops
Monitoring / on-call	€12,000	Engineering / SRE
Model re-evaluation	€7,500	3 changes × €2,500; AI lead

Cost line	Base €/year	Basis / owner
Expected errors	€19,848	Per category probability × cost × equivalents; Ops / CCO / DPO

Flip-point derivation. Base net value at tracing time-saving rate x = €26,437.54631 + €239,379.13125 × x. Solve against €125,000: x = 0.4117420478. Use **41.2%** as the conservative rounded requirement. A 10 percentage-point change is worth €23,938/year under this model.

This threshold depends on retaining all other full-scope assumptions. If disputes, compliance or strategic value are excluded, recalculate the business case; the S3 tracing-only pass cannot authorise the full investment.

Tracing saving	Net annual value	Difference vs €125k
30.0%	€98,251	€-26,749
41.1742%	€125,000	€0
41.2%	€125,062	€62
45.0%	€134,158	€9,158
48.0%	€141,340	€16,340
60.0%	€170,065	€45,065
65.0%	€182,034	€57,034
75.0%	€205,972	€80,972

Inputs — Volume and human cost

Input / Excel row	Low	Base	High	Source type	Note / validation
6: Volume per month (tickets, requests, conversations…) (tickets)	10500	12000	14000	Course scenario	12-month ticketing logs from Zendesk (144k tickets/ yr); low/high = seasonal dip/peak
7: Tier-1 agents (people)	16	18	22	Scenario assumption	Support operations headcount dedicated to merchant payments & disputes
8: Loaded cost per agent, incl. tooling (€ / year)	52000	55000	60000	analogy	Loaded salary for European fintech payment specialist + tooling seats (Zendesk, Slack, Jira)
9: Fully loaded cost of one unit handled by a human (€ / ticket)	8.5	10	11.5	analogy	Fully loaded cost per ticket handled by human (matches Paylio class benchmark and ~18 min AHT)

Inputs — Categories, eligibility, success and errors

Input / Excel row	Low	Base	High	Source type	Note / validation
14: Payment tracing & reconciliation ("Where is my money?") (share)	0.48	0.5	0.52	Scenario assumption	Ticketing tag "payment_tracing" (SEPA/SWIFT wire delays, payout reconciliation)
15: Disputes, chargebacks & fraud inquiries (share)	0.28	0.3	0.32	Scenario assumption	Ticketing tag "dispute_chargeback" (Visa/Mastercard disputes, evidence collection)
16: KYC, onboarding & PSD2 regulatory holds (share)	0.18	0.2	0.22	Scenario assumption	Ticketing tag "kyc_compliance" (SCA failures, payout reserve holds, IBAN re-verification)
18: Eligible share · Tracing (share)	0.65	0.75	0.82	guess	Guess; audit planned on 200 cases, not completed. Audit of 200 tickets: answerable from core transaction ledger and banking clearing logs
19: Eligible share · Disputes (share)	0.5	0.6	0.7	guess	Guess; audit planned on 200 cases, not completed. Audit of 200 tickets: card scheme rules, deadlines and evidence templates extractable by AI
20: Eligible share · KYC (share)	0.4	0.5	0.6	guess	Guess; audit planned on 200 cases, not completed. Audit of 200 tickets: compliance hold reasons and missing KYC docs extractable from DB flags

Input / Excel row	Low	Base	High	Source type	Note / validation
22: Time saving · Tracing (share)	0.3	0.48	0.58	guess	THE assumption that flips the decision: share of handling time saved by auto-drafting. Tested on 200 tickets in S4.
23: Time saving · Disputes (share)	0.2	0.32	0.42	guess	Share of handling time saved: auto-drafting dispute timeline and required proof. Tested in S4.
24: Time saving · KYC (share)	0.15	0.25	0.35	guess	Share of handling time saved: drafting explanation of regulatory hold and required KYC proof. Tested in S4.
26: Error rate · Tracing (share)	0.045	0.025	0.015	guess	From eval-set target (S5): incorrect clearing status or ledger transaction mismatch
27: Error rate · Disputes (share)	0.055	0.03	0.018	guess	From eval-set target (S5): incorrect dispute deadline advice or improper evidence checklist
28: Error rate · KYC (share)	0.04	0.02	0.012	guess	From eval-set target (S5): misidentifying compliance hold reason or citing wrong PSD2 article
30: Cost/error · Tracing (€ / error)	25	15	10	judgement	Merchant re-opens ticket + tier-2 banking ops re-investigation labor cost
31: Cost/error · Disputes (€ / error)	150	90	60	judgement	Lost chargeback dispute liability and card scheme filing penalty
32: Cost/error · KYC (€ / error)	250	150	90	judgement	Merchant escalation, legal review, and compliance remediation labor
34: Include · Tracing (1/0)	1	1	1	decision	In scope — primary time-sink for tier-1 agents
35: Include · Disputes (1/0)	1	1	1	decision	In scope — high value in drafting structured dispute responses
36: Include · KYC (1/0)	1	1	1	decision	In scope under L2 human approval only; never answers autonomously

Inputs — Adoption and variable AI cost

Input / Excel row	Low	Base	High	Source type	Note / validation
39: Q1 (share)	0.1	0.15	0.2	decision	Pilot on 3 senior payment specialists validating grounding evaluator
40: Q2 (share)	0.25	0.35	0.45	decision	Expanded to 8 agents; Cat 1 payment tracing fully deployed
41: Q3 (share)	0.45	0.6	0.7	decision	Rollout across all 18 agents for Cat 1 and Cat 2
42: Q4 (share)	0.6	0.75	0.85	decision	Mature adoption with continuous drift monitoring and L2 guardrails
45: AI cost per conversation (tokens, inference) (€ / conv.)	0.05	0.035	0.025	Scenario assumption	Planning allowance; published-rate calculation rechecked in price annex, not measured spend.
46: Retry / escalation overhead (×)	1.25	1.15	1.08	analogy	15% multi-step retrieval / context re-fetch overhead

Inputs — Fixed costs, comparisons and probabilities

Input / Excel row	Low	Base	High	Source type	Note / validation
49: Integration (CRM, knowledge base, auth, logging) — one-off (€)	75000	60000	45000	judgement	Zendesk sidebar plugin + internal payment ledger/ clearing API connectors + security audit
50: Amortisation period (years)	3	3	3	decision	3-year standard software amortisation
51: Evaluation — eval-set upkeep and human graders (€ / year)	30000	22000	16000	judgement	Golden dataset of 500 tickets + continuous LLM-as-a-judge grounding benchmark
52: Supervision — humans in the loop (L2) (FTE)	0.5	0.3	0.2	decision	Dedicated QA payment specialist auditing 5% of drafts and edge cases (FTE)
53: Cost per supervision FTE (€ / year)	55000	55000	55000	Scenario assumption	Loaded salary of payment QA specialist

Input / Excel row	Low	Base	High	Source type	Note / validation
54: Monitoring and on-call (€ / year)	18000	12000	8000	judgement	Observability stack (Datadog/LangSmith), latency alerts, on-call support
55: Model-version changes per year (changes)	4	3	2	analogy	Provider model updates (e.g. Anthropic/OpenAI minor/major releases)
56: Re-evaluation cost per model change (€ / change)	4000	2500	1800	judgement	Automated regression replay of 500-ticket eval suite + human spot check
59: Do-nothing: yearly efficiency gain of the current process (share)	0.03	0.04	0.05	analogy	Natural productivity gain (agents learn macros, documentation improves)
60: Next-best project — net annual value (the hurdle) (€ / year)	100000	125000	145000	judgement	Merchant Self-Serve Dispute & Reconciliation Portal (the hurdle)
63: Probability of each scenario (p)	0.2	0.6	0.2	judgement	Scenario probabilities (sum = 1.00)
67: SLA penalty avoidance on enterprise merchant contracts (€ / yr)	15000	35000	50000	judgement	35 enterprise SLA breaches avoided a year × €1,000 penalty per contract
68: Merchant churn retention on high-volume accounts (€ / yr)	12000	30000	45000	judgement	800 merchants × 3% at risk × 15% retained by faster dispute resolution × €8,000 margin

05 · Build, buy or orchestrate

Question	Weight	Build	Buy	Orchestrate	Reason (planning judgement)
Strategy — Is it our edge? Would a competitor buying the same tool erase our advantage?	0.1	3	1	4	Generic support bots cannot reconcile complex multi-leg B2B payment ledgers; orchestration integrates proprietary internal ledger APIs.
Strategy — Do we own data or know-how that no vendor has?	0.15	4	1	5	Transaction histories, SEPA/SWIFT clearing webhooks and merchant KYC data are internal; orchestration keeps banking data strictly inside our VPC.
Fit & speed — Does a product already do 80 % of the job — our languages, channels, integrations?	0.1	2	4	4	Foundation LLMs (Claude 3.5 Sonnet / GPT-4o) excel at financial reasoning; no need to train foundation base models from scratch.
Fit & speed — Must it be live in weeks, or can it take quarters?	0.1	1	5	4	Full scratch build takes 9–12 months; orchestrating existing agent frameworks + Zendesk plugin ships MVP in 6–8 weeks.
Fit & speed — Do we have the people to build it and to run it — evals, on-call, re-evaluation?	0.1	2	4	4	Team has strong ML/SWE talent for RAG orchestration and eval harnesses, but lacks bandwidth for custom model pretraining/hosting.
Money — 3-year total cost: fees per resolution or per seat vs the seven cost lines of a build	0.1	2	2	5	Vendor per-resolution pricing escalates to €100k+/yr at scale; orchestration keeps marginal token cost under €0.04 per conversation.
Money — At what volume does per-resolution pricing cost more than building? (break-even below)	0.05	3	2	4	At 144k annual tickets, per-resolution SaaS fees rapidly exceed the fixed amortised integration cost of an orchestrated copilot.
Risk & control — Where does our data go — GDPR, residency, may the vendor train on it?	0.15	5	1	4	B2B payment data requires strict PCI-DSS and GDPR residency; vendor multi-tenant clouds present severe compliance and data-leak risks.
Risk & control — Can we leave — exit clause, export of data, prompts and eval-sets?	0.05	5	1	4	Orchestration gives complete portability to switch foundation LLM providers (Anthropic, OpenAI, Mistral) without vendor lock-in.
Risk & control — Who is liable when it is wrong, and can we see why it answered — logs, evals?	0.1	4	1	5	We require custom Grounding Evaluator guardrails, full audit logs, and deterministic verification to prevent wrong transaction matches.

06 · Price check — 6 November 2026

Published-rate check, not a replacement for the workbook scenario. OpenAI GPT-4o mini model page lists standard text rates of **$0.15 / million input tokens** and **$0.60 / million output tokens**. The S2 note assumes 4,000 input + 450 output tokens across generator and checker: **$0.00087** per combined pass, or **$0.0010005** with 15% overhead, before extra calls, tooling and tax. This does not validate the proposed checker or production fit. The workbook's **€0.035** is retained as a conservative planning allowance, not a measured API price; no USD/EUR conversion is assumed.

Intercom public pricing states Fin **from $0.99 per outcome**. The historical workbook uses **€0.92 per resolution**, without a documented dated exchange rate. Keep it as a scenario input, not a current EUR quote. Platform fees, minimums and the exact outcome definition require a vendor quote. Recheck commercial terms and actual call counts before investment.

These checks do not retroactively assert that prices were checked on 16 October. The earlier source mentions of Claude 3.5 Sonnet / GPT-4o-mini are alternatives, not proof of one deployed provider. The published-rate illustration above uses GPT-4o mini only.

Build/buy interpretation and cost-of-error ownership

Weighted scores: Build 3.15, Buy 2.15, Orchestrate 4.35; weights sum to 100%. These are judgements, not vendor diligence. Internal comparable cost = €49,500 + €0.04025 × N; vendor scenario = €0.92 × N. Break-even ≈56,266 time-equivalent resolutions/year. At 17,489.16 equivalents, internal €50,204 vs vendor €16,090; buying is €34,114 cheaper on this limited comparison. Connector/platform fees, commercial terms and real billable outcomes must be checked; a copilot time-equivalent is not automatically a billable outcome.

CCO/DPO own disclosure and regulatory risk; Ops owns wrong answers/deadlines and oversight; Engineering owns authorization, exact checks and shutdown. €15/€90/€150 error severities price operational loss, not the catastrophic tail. Model confidence cannot authorize a match; any critical exposure disables drafts. Reporting duties require incident assessment.

Sources/Assignment_2_sources.zip retains the editable workbook, original memo and governance. Two genuine member-authored commits remain required. The PDF's one-page memo and recalculated annex supersede historical overclaims; original source limitations are explicitly disclosed.